Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 7: Weather, Climate and Adaptations of Animals to Climate

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	7 — Weather, Climate and Adaptations of Animals to Climate
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Watch for: treating one day's weather as proof of climate (and vice versa); reading a data table on time-of-extreme vs. value-of-extreme; confusing the cause of an adaptation with an incidental side effect; assuming a feature good for cold is also good for heat; mixing up element vs. instrument; and the surface-area-to-volume logic that runs opposite ways in polar vs. tropical animals. Do not write on this paper — mark answers on your answer sheet.

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1. A scientist has 40 years of daily temperature and rainfall records for a city and uses them to describe what that city is 'generally like'. The thing she is describing is the city's:

(A) weather, because temperature and rainfall are weather elements
(B) weather, because the records were taken on individual days
(C) altitude, because long records can only reveal height above sea level
(D) climate, because it is built from data averaged over a long span of years

2. Two statements are made about the same town: (i) 'It rained 9 mm here this morning.' (ii) 'This region is dry for most of the year.' Which is correct about these two statements?

(A) (i) describes weather and (ii) describes climate, and both can be true at once
(B) they contradict each other, so one of them must be false
(C) both describe weather, since both mention rain or dryness
(D) both describe climate, since both are about the same town

3. A traveller notices that on the day she visited, a normally rainy hill station was bright and sunny. She concludes the hill station's climate is sunny. The flaw in her reasoning is that she:

(A) ignored that sunshine is never part of weather at all

(B) forgot that climate changes from one day to the next

(C) used a single day's weather to judge a long-term climate

(D) assumed hill stations cannot have any sunshine ever

4. Place X has summers reaching 45 °C and winters near 2 °C every year; Place Y stays between 24 °C and 30 °C all year, every year. The most accurate statement is that:

(A) the two places differ only in the weather of the day each was measured

(B) Place X has no climate because its temperature keeps changing

(C) the two places differ in climate, since the contrast holds year after year

(D) both places have identical climates because both experience heat

5. Which question can only be answered by looking at climate data, not a single weather report?

(A) 'Which season is usually the wettest here?'(B) 'Is it raining right now outside?'

(C) 'What is today's maximum temperature?'

(D) 'How strong is the wind at this moment?'

6. A student says, 'Because the weather was different today and yesterday, the climate has clearly changed.' The best correction is:

(A) day-to-day differences are normal weather variation and do not by themselves show a change in climate

(B) the student is right, because two different days prove a climate shift

(C) the student is wrong because weather never changes from day to day

(D) the student is right, because climate is just yesterday's weather repeated

7. A weather report lists: temperature 29 °C, humidity 72%, rainfall 4 mm, wind 18 km/h, sunshine 6 hours. Which of these is measured with an anemometer?

(A) the rainfall of 4 mm (B) the humidity of 72%

(C) the temperature of 29 °C (D) the wind speed of 18 km/h

8. On a humid coastal day a hygrometer reads 88%. This number tells us the:

(A) depth of rain that fell, in millimetres

(B) amount of water vapour in the air, expressed as a percentage

(C) wind speed, in kilometres per hour (D) air temperature, in degrees Celsius

9. Why is rainfall reported in millimetres of depth rather than in litres collected?

(A) because rain gauges cannot hold any volume of water

(B) the depth of collected rain is independent of how wide the gauge's opening is, so it compares places fairly

(C) because millimetres measure how fast the rain fell, not how much

(D) because litres can only be used for measuring wind

10. A maximum-minimum thermometer is read once in the evening and shows a maximum of 33 °C and a minimum of 19 °C for that day. This single instrument is useful because it:

(A) measures the rainfall and the temperature at the same time

(B) shows only the temperature at the exact moment it is read

(C) records both the highest and lowest temperatures reached even when nobody was watching

(D) tells the wind direction along with the temperature

11. Which everyday observation is the most direct evidence that sunshine, not the time on the clock, is what drives daily heating?

(A) the clock shows a different number every minute

(B) the wind sometimes blows faster than at other times

(C) a spot in full sun feels much warmer than a shaded spot at the very same time of day

(D) rain gauges fill up during a storm

12. The Sun is called the ultimate source that drives weather because sunlight:

(A) directly blows the wind by pushing on the air with its rays

(B) is the only thing that makes thermometers exist

(C) cools the oceans down so that rain can form

(D) unevenly heats land, sea and air, and that heating sets winds, evaporation and clouds in motion

13. At night, with the Sun below the horizon, the air near the ground usually cools. The best explanation linked to the Sun is that:

(A) the Moon actively pulls heat out of the ground at night

(B) without incoming sunlight the ground stops being heated and loses its stored heat, so the air above cools

(C) wind always stops completely at night, removing all warmth

(D) humidity instantly drops to zero whenever the Sun sets

14. Places near the equator are generally hotter than places near the poles. The Sun-related reason is that near the equator the Sun's rays:

(A) are physically closer to the equator by thousands of kilometres

(B) shine for exactly 24 hours a day all year there

(C) carry more heat only because the equator has more cities

(D) strike more directly, concentrating their energy on a smaller area of ground

15. Which statement about the Sun and the seasons is correct?

(A) seasons arise from year-long changes in how sunlight reaches a place, an effect of Earth's tilt and orbit

(B) seasons happen because the Sun switches itself off in winter

(C) seasons change every few hours just like the weather

(D) seasons exist only because thermometers are read at different times

16. Why can a meteorologist forecast tomorrow's weather fairly well but not the weather on a specific date next year?

(A) the atmosphere has many interacting factors whose small differences grow rapidly, so far-off days cannot be pinned down

(B) because next year's Sun will be too weak to make any weather

(C) because climate, unlike weather, is impossible to study at all

(D) because thermometers stop working after a single day

17. A city's climate is described as 'hot summers, mild winters, heavy monsoon rain'. Which everyday plan should rely on this climate description rather than a daily forecast?

(A) deciding whether to carry an umbrella out this afternoon

(B) deciding in which months to schedule the city's main rainwater-harvesting work for the year

(C) deciding what to wear in the next hour

(D) deciding whether tonight's cricket match will be rained off

18. Climate is said to be more stable than weather because climate:

(A) is an average over many years, so a few unusual days barely move it

(B) is measured only once and then never changes by definition

(C) ignores rainfall and temperature, which are the changeable things

(D) is simply whatever the weather happens to be this week

19. Over four consecutive years a town's average annual rainfall was 800, 820, 790 and 810 mm. In one of those years a single freak storm dropped 200 mm in a day. This shows that:

(A) the climate changed completely in the storm year

(B) an extreme single day does not greatly disturb the long-term climate average

(C) the town has no climate because rainfall varied

(D) the storm proves the town is becoming a rainforest

20. Read this five-day record:

Day : Mon Tue Wed Thu Fri

Max (°C) : 31 28 35 33 30

Min (°C) : 20 21 19 25 22

On which day was the daily temperature range the largest?

(A) Thursday, because it had the highest minimum
(B) Tuesday, because its maximum was the lowest
(C) Monday, because its minimum was the lowest
(D) Wednesday, with a range of 16 °C

21. Using the same table (Max: Mon 31, Tue 28, Wed 35, Thu 33, Fri 30; Min: Mon 20, Tue 21, Wed 19, Thu 25, Fri 22), the coolest night, judged by the lowest minimum, was on:

(A) Thursday, because it had the highest minimum
(B) Wednesday, because it had the highest maximum
(C) Monday, because its minimum was 20 °C, the lowest
(D) Wednesday, with a minimum of 19 °C

22. A week's rainfall (mm) was: Mon 6, Tue 0, Wed 14, Thu 0, Fri 5, Sat 25, Sun 10. On how many days did it rain, and what was the total?

(A) 7 rainy days, total 60 mm
(B) 5 rainy days, total 50 mm
(C) 5 rainy days, total 60 mm
(D) 2 rainy days, total 25 mm

23. A humidity log reads: 05:00 → 92%, 09:00 → 80%, 13:00 → 58%, 17:00 → 66%, 21:00 → 85%. The reading at 13:00 is the lowest mainly because:

(A) early-afternoon heating from the Sun lets the warmer air hold moisture less tightly as relative humidity, so the percentage dips
(B) the rain gauge emptied itself at exactly 13:00
(C) the wind stopped blowing only at 13:00
(D) the thermometer was broken in the afternoon

24. Four cities report their warmest month and coolest month. Which set of figures points to a tropical-rainforest type of climate (consistently hot, little seasonal swing)?

(A) warmest month 44 °C, coolest month 6 °C
(B) warmest month 5 °C, coolest month -30 °C
(C) warmest month 30 °C, coolest month 27 °C
(D) warmest month 40 °C, coolest month 38 °C with almost no rain

25. From a four-year rainfall table a town shows: Year1 1900 mm, Year2 2100 mm, Year3 2000 mm, Year4 1950 mm, with rain spread across almost every month. The climate this best indicates is:

(A) a hot desert, because deserts get the most rain
(B) a polar climate, because the totals are large numbers
(C) a tropical-rainforest type, with high rainfall well spread through the year
(D) a dry climate, because rainfall changes a little each year

26. A data card lists an animal's habitat as 'temperature 25-30 °C year-round, humidity above 80%, rainfall over 2000 mm a year, dense tall trees'. An animal adapted with strong grasping hands and a long tail fits here because such a place is a:

(A) tropical rainforest, where climbing and swinging among tall wet trees is essential

(B) polar region, where grasping tails grip the ice

(C) hot desert, where long tails store water

(D) cold mountain snowfield, where tails balance on snow

27. The toucan's very large, light beak is best explained as an adaptation to rainforest life because it lets the bird:

(A) store body fat inside the beak for the cold season

(B) keep its head warm during freezing nights(C) dig deep burrows in dry sandy soil

(D) reach fruit on thin branches that could not bear the bird's full weight

28. In a dim, dense rainforest where animals cannot see one another far away, loud calls (as in howler monkeys) are favoured because sound:

(A) raises the body temperature of the caller in the cold

(B) drives away the heavy daily rainfall

(C) makes the animal's fur turn white for camouflage

(D) travels around tree trunks and foliage to reach others that vision cannot, suiting the cluttered habitat

29. The lion-tailed macaque lives in the Western Ghats rainforests. If its forest is cleared, the most direct threat is that the macaque:

(A) becomes too cold because rainforests provide the only warmth

(B) loses the tall connected trees it depends on for food and movement

(C) is forced to grow thick white fur to survive

(D) must learn to swim across oceans to find ice

30. A red-eyed tree frog rests motionless and green among leaves by day. The clearest reason this colouring is an adaptation rather than a coincidence is that it:

(A) helps the frog blend with green leaves and avoid being spotted by predators

(B) keeps the frog warm during long polar nights

(C) lets the frog reflect sunlight to stay cool in a desert

(D) allows the frog to glow and attract prey in darkness

31. A polar bear's coat has been studied and found to have two layers and to sit over a thick fat layer, with the bear also being a strong swimmer with wide paws. Which single statement correctly pairs a feature with its main job?

(A) the wide paws spread the bear's weight on snow and help it paddle while swimming

(B) the fat layer is mainly to make the bear float higher than its prey

(C) the two fur layers are mainly to keep the bear cool in summer heat

(D) swimming ability mainly helps the bear climb trees in the forest

32. Penguins huddle in dense groups during the coldest, windiest periods. The cause-and-effect reasoning behind this is that huddling:

(A) increases each bird's exposed surface so it can shed extra heat

(B) helps the birds catch fish more easily on the open ice

(C) cools the birds down because crowding traps cold air

(D) reduces the body surface each bird exposes to the cold, cutting heat loss for the group

33. Why does a polar bear have small ears and a short tail, while a rainforest animal might have large ears?

(A) small ears in the cold are only for hearing better in storms

(B) large ears in the warmth are only to look bigger to predators

(C) small extremities reduce heat loss in the cold, whereas large ears help shed heat in the warmth

(D) ear size has no link to temperature in any climate

34. A migratory bird like the Siberian crane flies thousands of kilometres each year. Migration is best understood as a way to:

(A) permanently change into a species suited to the cold

(B) avoid a harsh season in one place by moving to a region where food and conditions are better

(C) grow a thicker fur coat than animals that stay put

(D) stop needing to eat until the harsh season ends

35. Two birds face the same harsh northern winter. Bird P flies far south each autumn and returns each spring, while Bird Q stays put and grows a much denser coat of feathers. The most accurate conclusion is that:

(A) only Bird P is responding to the climate, while Bird Q's change has nothing to do with the seasons

(B) both birds are responding to the same seasonal climate, one by relocating and the other by changing its body

(C) Bird Q will gradually turn into Bird P's species simply by staying behind

(D) neither bird's behaviour has any connection to the climate of the region

36. Migratory birds often set off before the worst weather actually arrives. The most likely trigger that prompts this early departure is a change in:

(A) day length and dropping food supply, which signal the harsh season ahead

(B) the colour of a single tree's leaves only(C) the number of people watching the birds

(D) the price of grain in nearby markets

37. An animal is described as having a thick double fur coat, a compact rounded body, small ears, white colouring and a layer of fat. Placed in a tropical rainforest, this animal would most likely:

(A) overheat and stand out, because every feature is built to conserve heat and hide in snow

(B) thrive, because thick fur is useful in every climate

(C) be perfectly camouflaged against the green leaves

(D) lose heat too fast and feel cold in the rainforest

38. Why does a compact, rounded body help a polar animal but a long, slender body suit a warm-climate animal?

(A) a compact body loses heat faster, which is what cold animals need

(B) a compact body has less surface area for its volume, so it loses heat slowly, while a slender body loses heat quickly, which suits the warmth

(C) body shape changes how tall the animal looks but not its heat loss

(D) a slender body keeps a warm-climate animal warmer than a compact one

39. Which feature would be LEAST helpful to a penguin living in a polar climate?

(A) a thick layer of fat under the skin (B) the habit of huddling closely with others

(C) a long, thin, featherless tail with a large exposed surface

(D) a compact body with small flippers

40. A nature note claims, 'The polar bear is white so that sunlight will warm it.' The correct response is that the white coat:

(A) is correct, since white absorbs the most heat from the Sun

(B) is white only to signal to other bears across the ice

(C) is white so prey can easily see it coming

(D) mainly camouflages the bear against snow and ice while hunting, rather than serving to warm it by colour

41. An exam answer says, 'Migration proves climate and weather are the same, since both involve travelling animals.' The flaw is that:

(A) migration has nothing to do with climate or weather at all

(B) weather and climate really are identical, so the answer is right

(C) migration depends only on one day's weather, not the seasons

(D) animals migrate in response to the long-term seasonal climate pattern, which is different from day-to-day weather

42. A barometer reading falling steadily through the morning is often used by forecasters as a hint that:

(A) unsettled or rainy weather may be approaching, since pressure changes precede many weather changes

(B) the climate of the place has permanently changed that morning

(C) the amount of rainfall already collected is rising

(D) the wind has stopped blowing entirely

43. A six-day table of daily maximum temperatures reads 28, 27, 29, 28, 27, 28 °C, while a desert town the same week reads 41, 18, 43, 16, 40, 17 °C. Comparing the two patterns, the best statement is:

(A) both places have identical climates because their averages are similar

(B) the first place has small day-to-day swings while the desert has very large ones, a difference rooted in their climates

(C) the desert has no climate because its temperature jumps around

(D) the first place's steadiness proves it has no weather at all

44. Two friends compare notes: one says 'My city is humid', the other says 'My city is humid today'. The difference between these statements is that the first describes _____ and the second describes _____.

(A) weather; climate | (B) climate; weather

(C) weather; weather | (D) altitude; humidity

45. A rain gauge with a 200 mm-deep collecting jar overflows after a storm, and the observer records 'over 200 mm'. The main lesson for accurate weather data is that:

(A) an instrument can only record within its limits, so an overflowing gauge gives only a minimum, not the true total

(B) the storm must have produced exactly 200 mm of rain

(C) rain gauges measure wind once they overflow

(D) an overflow means no rain actually fell

46. Why do tropical rainforests support such a great variety of animals compared with polar regions?

(A) because the polar cold makes animals reproduce far faster

(B) their warm, wet, stable climate provides abundant food and shelter all year, supporting many kinds of life

(C) because rainforests have almost no plants for animals to use

(D) because variety depends only on how many people visit a place

47. During a long polar winter the Sun may not rise for weeks. The most direct consequence for the climate there is that:

(A) the missing sunlight makes the region suddenly hot

(B) with little or no incoming sunlight for long periods, temperatures stay extremely low

(C) rainfall becomes the heaviest on Earth during that time

(D) the lack of Sun has no effect on temperature at all

48. An elephant in a tropical region has very large ears that it often flaps. Linking feature to climate, the large ears most likely help the elephant to:

(A) trap heat to keep warm in freezing weather

(B) hear sounds coming from under the snow(C) store fat for a long cold winter

(D) lose excess body heat in the warm climate by exposing a large surface and flapping air over it

49. A polar bear can stay warm in icy water far longer than an unprotected animal could. The combination most responsible is its:

(A) large thin ears and long bare tail (B) bright camouflage colour and loud calls

(C) thick fat layer plus dense water-shedding fur, which together greatly slow heat loss

(D) grasping hands and a long climbing tail

50. A weather summary states: 'Maximum 7 °C, minimum -12 °C, light snow, 20 hours of darkness.' This single report most strongly suggests the place has a:

(A) polar-type climate, given the freezing temperatures, snow and very long night

(B) tropical-rainforest climate, because there is precipitation

(C) hot-desert climate, because the temperature range is wide

(D) warm coastal climate, because the maximum is above zero

51. Why is it misleading to say 'It is very cold today, so this must be a polar region'?

(A) because cold days never occur outside polar regions

(B) because a single cold day actually proves a polar climate

(C) one cold day is weather and could happen in many places, while a polar region is defined by a cold pattern lasting most of the year

(D) because temperature has nothing to do with defining a polar region

52. A table compares two habitats: Habitat A averages 27 °C with 2200 mm rain a year; Habitat B averages -20 °C with 150 mm precipitation a year. An animal with two fur layers, a fat store and white colour belongs in:

(A) Habitat A, because thick fur is needed against the heavy rain

(B) Habitat A, because white colour blends with green leaves

(C) Habitat B, whose cold, icy conditions match heat-conserving, snow-camouflaged features

(D) either habitat equally, since fur works the same everywhere

53. Which reasoning correctly connects the Sun to the wind that appears in a weather report?

(A) the Sun's rays physically shove the air sideways to make wind

(B) wind is made by the Moon and has no link to the Sun

(C) wind appears only because anemometers are switched on

(D) the Sun heats different areas unequally, creating temperature and pressure differences that set the air moving as wind

54. A child argues, 'Forecasts are useless because they only work for a few days.' A fair scientific reply is:

(A) the child is right, because no forecast is ever accurate

(B) forecasts fail only because the Sun is unpredictable

(C) forecasts could easily cover a whole year if more thermometers were used

(D) short-range forecasts are genuinely useful precisely because the atmosphere is too changeable to predict far ahead, so they make the most of what can be known

55. A migrating flock returns to the same northern wetland each spring to breed and flies south each autumn. This yearly back-and-forth is governed by:

(A) the random weather of a single chosen day

(B) the regular seasonal climate cycle, which makes each region favourable at different times of year

(C) the birds permanently settling and never returning

(D) a sudden one-time change that never repeats

56. Among these, the clearest case of a structure whose form is explained by a HOT-WET climate (and would be a disadvantage in the cold) is:

(A) a thick fat layer under the skin (B) a compact body with very small ears

(C) long limbs and a grasping tail for climbing dense, wet rainforest trees

(D) dense white fur that sheds water

57. A six-day record of sunshine hours is: Mon 8, Tue 2, Wed 9, Thu 1, Fri 7, Sat 0. Which conclusion is best supported, and what does it illustrate?

(A) sunshine was the same every day, illustrating stable climate

(B) the totals prove the region's climate changed within the week

(C) sunshine varied sharply from day to day, illustrating how weather can change quickly

(D) the figures show the Sun shone for the same hours regardless of clouds

58. An animal expert states: 'Adaptations are features that, over long times, suit an animal to its climate.' Which observation best fits this definition?

(A) a single bear that happened to feel warm on one mild day

(B) a monkey that learns a new trick during an afternoon

(C) polar bears born generation after generation with thick fur and fat, matching the long-cold polar climate

(D) a bird that flies into a warm house for one night

59. A region that is hot and humid with rain almost daily, dense tall trees, and animals like toucans, tree frogs and lion-tailed macaques is most precisely classed as:

(A) a polar climate region (B) a tropical rainforest climate region

(C) a hot desert climate region (D) a cold mountain snowfield region

60. Which final statement about weather and climate is fully correct?

(A) climate is short-term and weather is the many-year average

(B) weather and climate are two names for exactly the same measurement

(C) weather is measured by instruments but climate cannot be measured in any way

(D) weather describes the short-term state of the atmosphere at a place, while climate describes the average of that weather over many years

Solutions — Science (Class 7), Chapter 7: Weather, Climate and Adaptations of Animals to Climate — Paper 3

Paper 3 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	C	21	D	31	A	41	D	51	C
2	A	12	D	22	C	32	D	42	A	52	C
3	C	13	B	23	A	33	C	43	B	53	D
4	C	14	D	24	C	34	B	44	B	54	D
5	A	15	A	25	C	35	B	45	A	55	B
6	A	16	A	26	A	36	A	46	B	56	C
7	D	17	B	27	D	37	A	47	B	57	C
8	B	18	A	28	D	38	B	48	D	58	C
9	B	19	B	29	B	39	C	49	C	59	B
10	C	20	D	30	A	40	D	50	A	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Climate is the average pattern of the atmosphere worked out over many years. Even though each individual record is a day's weather, summarising 40 years into 'what the place is generally like' produces climate. The first wrong choice confuses the raw measurements (weather elements) with the long-term summary they build into. The next assumes that because each reading is a day's weather, the whole 40-year picture stays weather, which ignores the averaging. The last wrongly claims long records reveal altitude; altitude is a fixed geographic property, not something derived from a temperature-rainfall series.

2. (A) Statement (i) is a single morning's event, which is weather; statement (ii) is the long-term average pattern, which is climate. A dry region can still have a rainy morning, so the two coexist without contradiction. The first wrong choice assumes a wet morning must clash with an overall dry climate, but one day cannot overturn a yearly average. The second ignores that (ii) spans 'most of the year', a climate span, not a moment. The third ignores that (i) is plainly a single short event, the very definition of weather.

3. (C) Climate is the multi-year average, so one sunny day says nothing reliable about it; that single-day-to-climate jump is the flaw. The first wrong choice is false because sunshine is a recognised weather element. The second is false because it is weather, not climate, that varies day to day; climate is precisely the stable long-term picture. The third misstates her error: she did not deny sunshine exists, she over-generalised one day of it.

4. (C) Because the wide swing at X and the steadiness at Y repeat 'every year', the difference is a climate difference, not a one-day fluke. The first wrong choice reduces a yearly repeated pattern to a single day's weather. The second is wrong because a large seasonal swing is itself a climate feature, not an absence of climate. The third ignores that the actual yearly patterns are clearly different, so 'identical' fails despite both being warm at times.

5. (A) 'Usually the wettest season' asks for a pattern that recurs across years, which is a climate question. The other three each ask about an instantaneous or single-day state, which a single weather report answers. 'Raining right now' is the present moment; 'today's maximum temperature' is one day; 'wind at this moment' is an instant, so none of them needs multi-year data.

6. (A) Weather naturally varies from one day to the next; climate is the long-term average, which two days cannot establish or overturn. The first wrong choice accepts an invalid two-day proof. The second falsely claims weather is constant day to day, which is the opposite of reality. The third wrongly equates climate with a single previous day, when climate is a many-year average.

7. (D) An anemometer measures wind speed, so the 18 km/h figure is its reading. Rainfall is measured by a rain gauge, temperature by a thermometer, and humidity by a hygrometer, so the other three figures come from different instruments and are not anemometer readings.

8. (B) A hygrometer measures humidity, the water vapour in the air, reported as a percentage; 88% means the air holds a high share of moisture. Rainfall depth in millimetres comes from a rain gauge, wind speed from an anemometer, and temperature from a thermometer, so none of those three is what a hygrometer's 88% represents.

9. (B) A wider gauge catches more litres but the rain still piles up to the same depth, so depth (mm) lets different stations be compared regardless of opening size. The first wrong choice falsely claims gauges hold no volume; they do collect water, which is then read as a depth. The second confuses depth (an amount) with rate (speed of fall). The third invents a wind link that does not exist; litres are a volume, not a wind measure.

10. (C) The maximum-minimum thermometer captures the day's extreme high and low automatically, so one evening reading reveals both without constant monitoring. The first wrong choice adds rainfall, which is a rain gauge's job. The second describes an ordinary thermometer showing only the present reading, missing the whole point of the max-min design. The third invents a wind-direction function the instrument does not have.

11. (C) If two spots at the same clock time differ in warmth only because one is sunlit, the heating clearly tracks sunlight, not the clock. The first wrong choice notes the clock changes but says nothing about heat. The second is about wind, which does not show sunlight is the heat source. The third concerns rainfall collection, unrelated to whether sun or clock controls warming.

12. (D) Solar heating is uneven, and the resulting temperature differences drive the winds, evaporation and cloud formation that make up weather, so the Sun is the root energy source. The first wrong choice imagines sunlight mechanically pushing air; in fact it works by heating, which then causes air to move. The second confuses the Sun's role with the existence of instruments. The third reverses the physics: the Sun warms and evaporates ocean water rather than cooling it to make rain.

13. (B) Once the Sun sets, the ground no longer receives solar energy and radiates away the heat it stored, so the air cools, showing how solar input governs temperature. The first wrong choice gives the Moon an active cooling power it does not have. The second falsely claims wind stops every night. The third makes humidity vanish at sunset, which is untrue and is not the reason for cooling.

14. (D) Near the equator sunlight hits the ground more head-on, so the same beam of energy is concentrated over a smaller patch, producing more intense heating than the slanted rays at the poles. The first wrong choice misjudges distance; the tiny difference in distance to the Sun is negligible compared with the angle of the rays. The second falsely claims 24-hour daylight year-round at the equator. The third wrongly attributes heat to the number of cities.

15. (A) Seasons come from the steady, year-long variation in how much and how directly sunlight reaches a place as Earth tilts and orbits, a climate-scale solar effect. The first wrong choice is false because the Sun does not turn off. The second confuses seasons (which span months) with weather (which changes hourly). The third absurdly blames the timing of thermometer readings rather than the actual solar geometry.

16. (A) Weather is a fast-changing system of many interacting factors; tiny present uncertainties balloon over time, so short-range forecasts work while a specific distant date does not. The first wrong choice falsely weakens next year's Sun. The second wrongly says climate cannot be studied, when in fact long-term averages are well studied even though specific far-off days are not. The third invents a one-day instrument failure.

17. (B) Choosing the months for rainwater harvesting depends on the recurring yearly pattern, which is exactly what climate describes. The other three are immediate, single-occasion decisions that depend on the day's forecast: an umbrella this afternoon, clothing in the next hour, and tonight's match are all weather questions, not climate questions.

18. (A) Averaging over many years smooths out individual odd days, so climate shifts only slowly, which is why it seems stable. The first wrong choice claims climate is fixed by a single measurement, but it is a continually updated long-term average. The second wrongly says climate ignores temperature and rainfall, when those are precisely the elements averaged into it. The third equates climate with a single week, far too short to be a climate.

19. (B) The yearly averages stayed close to 800 mm despite the freak storm, demonstrating that one extreme day barely moves the multi-year climate average. The first wrong choice claims a total climate change, but the numbers stayed similar. The second denies a climate exists merely because of slight variation, yet a stable ~800 mm pattern is itself a climate. The third leaps to a rainforest from one storm, ignoring the steady yearly totals.

20. (D) The range is max minus min: Mon 31-20=11, Tue 28-21=7, Wed 35-19=16, Thu 33-25=8, Fri 30-22=8. Wednesday's 16 °C is clearly the largest. Thursday is wrong: a high minimum actually narrows the range, giving only 8 °C. Tuesday's low maximum pairs with a high minimum for the smallest range of 7 °C, not the largest. Monday's low minimum gives 11 °C, less than Wednesday's, so it is not the largest.

21. (D) The coolest night is the lowest minimum: Wed 19 °C is the lowest of 20, 21, 19, 25, 22. Thursday is wrong because 25 °C is the warmest night, not the coolest. The second Wednesday option gives a true day (Wednesday) but the wrong reason: a high maximum does not define the coolest night. Monday's 20 °C is low but not the lowest, since Wednesday's 19 °C beats it.

22. (C) Days with rain are those above 0: Mon, Wed, Fri, Sat, Sun, which is 5 days; the total is 6+0+14+0+5+25+10 = 60 mm. The first wrong choice counts all 7 days as rainy, but Tuesday and Thursday recorded 0. The second has the right number of rainy days but mis-adds the total to 50. The third counts only the two biggest days and ignores the smaller rain amounts.

23. (A) Around early afternoon the Sun has heated the air the most, and warm air drives the relative humidity percentage down, so 58% at 13:00 is the lowest of the day. The first wrong choice invents a rain-gauge event unrelated to humidity. The second blames wind, which does not set humidity in this way. The third blames a broken thermometer, but the humidity values come from a hygrometer and follow the expected daytime dip.

24. (C) A rainforest climate stays hot all year with a small swing, matching 30 °C down to 27 °C (and, by reputation, heavy rain). The first wrong choice has a huge 38-degree swing, typical of an inland temperate or desert-edge climate, not a rainforest. The second is far too cold, fitting a polar climate. The third stays hot but is described as almost rainless, which is a hot desert, not a wet rainforest.

25. (C) Roughly 2000 mm spread across nearly every month is the signature of a hot, wet rainforest climate. The first wrong choice contradicts itself: deserts are defined by very little rain, not the most. The second confuses a large rainfall number with cold polar conditions, which actually have very low precipitation. The third calls it dry despite about 2000 mm a year, mistaking small year-to-year wobble for dryness.

26. (A) The hot, humid, heavily forested figures describe a tropical rainforest, where grasping hands and a long tail are used to move through the trees. The first wrong choice puts a tree-climbing adaptation on bare ice where there are no trees to grasp. The second invents water storage in a tail, which is not how desert animals work. The third places dense-forest climbing gear on a treeless snowfield.

27. (D) In dense forest, much fruit hangs on slender branches, and a long light beak lets the toucan reach it without venturing onto branches too weak to hold it. The first wrong choice imagines fat storage in a beak, which is not its function. The second gives a cold-climate purpose to a bird of the warm rainforest. The third describes desert burrowing, irrelevant to a tree-living fruit-eater.

28. (D) Thick vegetation blocks sightlines, so calls that carry past trunks and leaves are the practical way to communicate in a rainforest, explaining loud-calling animals. The first wrong choice treats calling as a heating mechanism, which it is not, and the rainforest is warm anyway. The second imagines sound repelling rain. The third links sound to fur colour change, an unrelated and false idea.

29. (B) A rainforest macaque is tied to its canopy for both food and travel, so clearing the trees directly removes what it needs to survive. The first wrong choice misframes the threat as cold, but the danger is lost habitat, not temperature. The second imagines a sudden growth of polar fur, which does not happen. The third invents an ocean-and-ice scenario unrelated to forest loss.

30. (A) Green against green leaves is camouflage that reduces the chance of a predator spotting the still frog, a clear survival benefit in its leafy habitat. The first wrong choice gives a cold-climate warming role to a colour, which colour cannot provide, and the frog lives in warmth. The second invents a desert-cooling function. The third claims the frog glows to lure prey, which green leaf-camouflage does not do.

31. (A) Wide paws act like snowshoes on soft snow and like paddles in water, so that feature-job pairing is correct. The first wrong choice gives the fat a flotation purpose, but its real jobs are insulation and energy storage. The second claims the fur cools the bear, when fur insulates against extreme cold. The third ties swimming to tree climbing, which is irrelevant to a treeless polar habitat.

32. (D) Packing together hides much of each penguin's surface inside the group, so less skin faces the cold and the whole huddle loses heat more slowly. The first wrong choice reverses the logic: huddling decreases, not increases, exposed surface. The second links huddling to fishing, which happens in water, not in a land huddle. The third claims huddling cools them, the opposite of its warming purpose.

33. (C) Extremities lose heat fast, so a polar animal benefits from small ones to conserve warmth, while a warm-climate animal benefits from large ones to release heat, exactly opposite needs set by climate. The first wrong choice replaces the heat reason with hearing. The second replaces the heat-shedding reason with looking big. The third denies any temperature link, contradicting the clear pattern across climates.

34. (B) Migration lets an animal sidestep a harsh season by relocating to where conditions and food are favourable, then returning later. The first wrong choice confuses moving with turning into a new species, which does not happen. The second describes growing fur, a different strategy used by animals that stay, not by migrating birds. The third imagines the bird fasting through the season instead of travelling.

35. (B) Both birds are coping with the same harsh seasonal climate: Bird P relocates (migration) while Bird Q changes its body by growing a denser coat, two different responses to one seasonal cue. The first wrong choice denies that Bird Q's thicker coat is climate-driven, but growing winter plumage is a clear seasonal response. The third imagines one species turning into another merely by staying, which does not happen. The fourth severs both behaviours from the climate, contradicting that each is a seasonal response.

36. (A) Shortening days and a falling food supply are reliable advance cues of an approaching harsh season, so birds use them to leave in good time. The first wrong choice fixes on one tree's leaf colour, far too local to be a dependable signal. The second and third tie a natural behaviour to human crowd size and market prices, neither of which a bird perceives or responds to.

37. (A) All those features minimise heat loss and provide snow camouflage, so in a hot, green rainforest the animal would overheat and its white coat would be conspicuous, not hidden. The first wrong choice claims thick fur suits every climate, but it is a liability in heat. The second says white camouflages against green, which it plainly does not. The third claims it would lose heat too fast, the opposite of what its heat-conserving build would do in warmth.

38. (B) Less surface area per unit of volume means slower heat loss, helping in the cold, whereas more surface area sheds heat faster, helping in warmth, so the two body shapes match opposite climates. The first wrong choice reverses the cold logic. The second denies any heat effect of shape, contradicting the surface-area principle. The third claims a slender body keeps an animal warmer, when in fact it cools faster.

39. (C) A long, thin, bare tail with lots of exposed surface would shed precious body heat, making it the least helpful feature in the cold. A fat layer insulates and stores energy, huddling cuts group heat loss, and a compact body with small flippers limits surface area, so all three of those genuinely help a polar penguin.

40. (D) The white coat's chief benefit is blending with snow so the bear can approach prey unseen; warming is provided by fur and fat insulation, not by the colour. The first wrong choice claims white absorbs the most heat, which is the opposite of how light colours behave and is not the coat's purpose. The second reduces camouflage to mere signalling. The third would make camouflage useless by letting prey spot the bear.

41. (D) Birds migrate to track recurring seasonal conditions, the climate pattern, which is distinct from the day-to-day weather, so the two are not the same. The first wrong choice denies any climate link, but migration is clearly climate-driven. The second accepts the false claim that the two terms are identical. The third reduces migration to a single day's weather, ignoring the seasonal, multi-month basis of the journey.

42. (A) A barometer measures air pressure, and a falling reading often precedes unsettled, rainy weather, which is why forecasters watch it. The first wrong choice leaps from a morning reading to a permanent climate change, but climate is a multi-year average. The second confuses pressure with collected rainfall, which a rain gauge measures. The third claims the wind stopped, which a barometer does not indicate and which falling pressure does not imply.

43. (B) The first town's readings barely move while the desert swings wildly between hot days and cold nights, and such persistent differences in variability reflect the two places' climates. The first wrong choice fixes on a rough average match but ignores the very different spreads. The second denies the desert a climate merely because it varies, yet large daily swings are themselves a climate feature. The third claims steadiness means no weather, but each daily reading is still weather.

44. (B) 'My city is humid' (in general) is a standing, long-term description, which is climate, while 'humid today' is a single day's state, which is weather. The first wrong choice swaps the two. The second labels both as weather, but the general statement is a climate description. The last invents altitude and a bare humidity label that do not match either statement's meaning.

45. (A) Once the jar overflows, the true rainfall exceeded its capacity, so the reading is only a lower bound, teaching that instruments report only within their range. The first wrong choice treats the brim level as the exact total, ignoring the unmeasured overflow. The second invents a switch to wind measurement. The third bizarrely concludes no rain fell, contradicting an overflowing gauge.

46. (B) Year-round warmth, moisture and plentiful plant growth give rainforests rich, steady food and shelter, allowing many species to coexist. The first wrong choice claims cold boosts reproduction, but the harsh polar climate actually limits life. The second falsely says rainforests lack plants, when they are densely vegetated. The third ties biodiversity to tourist numbers, which has nothing to do with it.

47. (B) Because solar energy is the main source of warmth, weeks without sunrise leave the polar region intensely cold. The first wrong choice claims absent sunlight causes heat, which reverses cause and effect. The second imagines record rainfall, but polar regions are dry and frozen. The third denies any temperature effect, contradicting the central role of the Sun in heating.

48. (D) Big, well-supplied ears expose a large surface, and flapping moves air over them to release heat, which suits a hot climate. The first wrong choice reverses this into heat trapping for cold, the opposite need. The second invents under-snow hearing, irrelevant to a warm habitat. The third treats ears as fat-storage organs for winter, which they are not.

49. (C) A deep fat layer and dense fur that sheds water work together to keep heat in even in freezing water, explaining the bear's endurance. The first wrong choice lists large thin ears and a long bare tail, which would lose heat, not conserve it. The second lists colour and calls, which do nothing for warmth in water. The third lists rainforest climbing features that have no role in polar swimming.

50. (A) Sub-zero temperatures, snow and a 20-hour night together point clearly to a polar climate. The first wrong choice seizes on 'precipitation' but snow at -12 °C is the opposite of a hot, wet rainforest. The second notes a wide range but ignores that both ends are cold or near-freezing, unlike a baking desert. The third clings to a 7 °C maximum as 'warm', but 7 °C with snow and long darkness is far from a warm coast.

51. (C) A polar region is defined by long-term, year-round extreme cold (climate), whereas a single cold day is weather that can occur in many non-polar places, so the inference is unsound. The first wrong choice falsely claims cold days happen only at the poles. The second accepts a single day as proof, the very error being exposed. The third denies that temperature defines polar regions, when persistent cold is exactly what defines them.

52. (C) Two fur layers, fat and white colour are heat-conserving and snow-matching, so they fit Habitat B's freezing, snowy climate. The first wrong choice treats fur as rain protection for the warm, wet Habitat A, where it would cause overheating. The second claims white blends with green leaves, which it does not. The third says fur suits any habitat equally, ignoring that thick fur is harmful in a hot climate.

53. (D) Uneven solar heating produces differences in temperature and pressure, and air moving in response is the wind, so the Sun is the underlying driver. The first wrong choice imagines rays pushing air directly rather than working through heating. The second assigns wind to the Moon and severs the Sun link, which is false. The third confuses measuring wind with causing it; the anemometer only records what the Sun-driven process creates.

54. (D) The limited range of forecasts reflects the atmosphere's rapidly growing uncertainty, yet within a few days they are reliable and valuable, so they are far from useless. The first wrong choice claims no forecast is ever accurate, contradicting everyday reliable short-range forecasts. The second blames an 'unpredictable Sun', when it is the complex atmosphere, not the Sun, that limits range. The third assumes more thermometers could extend forecasts to a year, ignoring the fundamental growth of uncertainty.

55. (B) The repeated yearly journey tracks the seasonal climate cycle, with north favourable in summer and south in winter, so the round trip recurs with the seasons. The first wrong choice attributes a yearly pattern to one random day's weather. The second denies the return that the question explicitly states. The third calls it a one-off change, contradicting the described annual repetition.

56. (C) Climbing limbs and a grasping tail suit life in tall wet forest and offer little protection against cold, so they are the hot-wet adaptation. The other three are all cold-climate features: a fat layer insulates, a compact small-eared body conserves heat, and dense white water-shedding fur both insulates and camouflages in snow, none of which are hot-wet adaptations.

57. (C) The hours jump from 8 to 2 to 9 and down to 0, a sharp day-to-day swing that illustrates how fast weather changes. The first wrong choice claims the values were equal, which the numbers contradict. The second reads a one-week swing as a climate change, but climate needs years, not a week. The third claims constant sunshine regardless of clouds, contradicting the large daily variation the data show.

58. (C) Inherited thick fur and fat appearing generation after generation in a long-cold climate is exactly a climate-matched adaptation built up over long times. The first wrong choice is a one-day comfort, not an inherited feature. The second is a learned afternoon trick, not a long-term climate adaptation. The third is a single night's behaviour, again neither inherited nor climate-scaled.

59. (B) Daily rain, heat, dense tall trees and that line-up of canopy animals together pin the region as a tropical rainforest. A polar region is freezing and treeless, a hot desert is dry with little vegetation, and a cold mountain snowfield is icy and bare, so none of those three matches the hot, wet, richly forested description.

60. (D) Weather is the short-term atmospheric state and climate is its long-term, many-year average, which is the standard and correct distinction. The first wrong choice swaps the two definitions. The second claims they are identical, ignoring the time scale that separates them. The third says climate cannot be measured, but climate is computed precisely from many years of measured weather data.